Thermodynamic Answer Key

Part 1:

1. -40
2. -5℉
3. B = AK, where A is Avogadro's constant ($6.022 * 10^{23}$)
4. 8.314 J / (mol·K)
5. A.

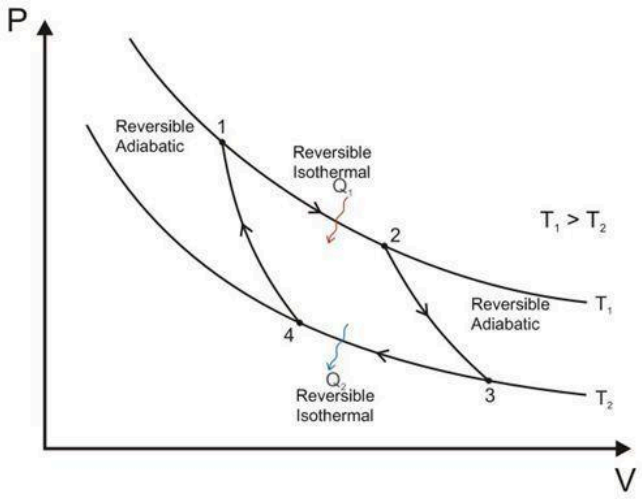
Part 2:

1. 326.36 kJ/kG
2. B
3. 14,981 seconds
4.
    a. 4Rln(2)
    b. The calculated entropy change should be the same, but physically adding or removing the barrier between the two chambers should not change the system, and therefore the entropy change should be zero.

        (This question should be graded for cogency, not accuracy. As long as they indicated that the entropy change should be zero and/or the calculated entropy change is false, give them full credit. Assume some width in reasoning - this is supposed to make them ask questions. If you have questions about grading, ask Surya)

5. Root mean square speed < mean speed < most probable speed

Part 3:



1. 10 pts
   - 5 pts for correct shape
   - 3 pts for labeled axes
   - 2 pts for labeled stages (1,2,3,4)

2. 10 pts (5 pts for each)

   $\Delta U = \Delta W = n \cdot C \cdot \Delta T$
   - Step 2 ($2 \rightarrow 3$): ΔW = 1 * 12.5 * 32 = 400 J
   - Step 4 ($3 \rightarrow 4$): ΔW = 1 * 12.5 * -32 = -400 J

   (Incorrect if wrong sign: negative instead of positive, etc)

3. The internal energy does not change after one cycle. State of the system is exactly the same as it started

   10 pts if right answer + explanation,
   5 pts if just answer and no description

4. (5 pts each)

   - Step 1 ($1 \rightarrow 2$): Heat added to system
   - Step 3 ($3 \rightarrow 4$): Heat removed from system

5. Carnot efficiency: $E = (T_H - T_C) / T_H = 32/432 = 0.074$

Part 4:

1. Gibbs
2. Rankine
3. Clausius
4. True
5. A